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Psychiatry in transition: an assessment in the context of societal change

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Abstract

Aims To explore how global challenges such as climate change, artificial intelligence (AI), and migration intersect with generational change among psychiatric trainees and reshape specialist training.

Methods An integrative review drawing on symposium contributions and a comprehensive literature review to assess the evolving priorities and challenges in psychiatric training.

Results Younger psychiatrists increasingly prioritize work-life balance, sustainability, and participatory learning environments. However, gaps remain in integrating climate-related knowledge, transcultural competence, AI literacy and neurodevelopmental disorders within psychiatric curricula. Generational tensions and traditional hierarchical structures further complicate training.

Conclusions A new training model is needed that fosters mutual understanding between generations, encourages reflective dialogue, and supports collaborative learning. Preparing psychiatrists for the future requires updated content and a commitment to relational transformation and co-created educational practices.

Introduction

Comparing the experiences of psychiatry trainees in different German-speaking countries throughout Europe, Wedmann et al. [103] described how the young generation of psychiatrists deliberately aligns their career decisions and needs with a sustainable work-life balance that supports both mental and physical health. The authors

addressed developmental scenarios in specialist training and highlighted system-relevant differences from a generational perspective.

Following this analysis, the question arises to what extent universal developments of the twenty-first century—such as climate change, artificial intelligence, and migration—are impacting the field of psychiatry and psychotherapy. Due to being closely interwoven with these global phenomena it is becoming increasingly evident that health care policies in many countries are initiating substantial changes in psychiatric care.

Psychiatric disorders are complex, dynamic conditions whose prevalence and manifestations are strongly influenced by societal systems and their upheaval.

This article explores how global transformation processes—such as climate change (planetary health), artificial intelligence, and migration—are shaping the content, structure, and political context of psychiatry and psychiatric specialty training. A focus is placed on the changing

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expectations of psychiatry trainees, who increasingly prioritize a healthy work-life balance and sustainable working conditions.

In addition the review explores how neurodevelopmental disorders (e.g. ADHD and autism spectrum disorders), which frequently persist into adulthood and intersect with psychosocial vulnerabilities, remain insufficiently integrated into adult psychiatry training curricula.

To address these questions we conducted a narrative integrative review combining insights from recent international symposia including the World Congress of Psychiatry 2023 (WCP 2023) and German Society for Psychiatry, Psychotherapy and Psychosomatics (DGPPN) Congress 2024, with a comprehensive literature search. This approach aimed to identify key themes shaping psychiatric specialist training rather than to quantify prevalence.

Based on preliminary observations and existing literature, we formulated the following hypotheses:

- (1) Global transformation processes—such as climate change, artificial intelligence, and migration—are increasingly influencing the content and structure of psychiatric training.
- (2) Generational change among psychiatrists interacts with these developments, creating tensions between traditional hierarchical models and emerging values such as sustainability and work-life balance.
- (3) Despite growing awareness, integration of climate-related knowledge, transcultural competence, AI literacy, and of certain neurodevelopmental disorders into psychiatric curricula remains limited.

Methods

Approach and literature search

Given the interdisciplinary nature of this topic we adopted a narrative integrative review approach combining insights from symposia presented at the World Congress of Psychiatry 2023 and DGPPN Congress 2024 with a broad literature search. The aim was to identify key themes shaping psychiatric specialist training rather than to quantify prevalence.

Search strategy

We searched PubMed, PsycINFO, and Google Scholar from inception to September 2025 without restrictions on language, country of origin, or publication date. Search terms included: “climate change,” “global warming,” “extreme weather,” “heat wave(s),” “psychiatry,” “psychotherapy,” “climate anxiety,” “eco-anxiety,” “psychiatric trainees,” “psychiatric wards,” “migration,” “brain drain,” “artificial intelligence,” “AI,” “specialization,” “mental health,” “mental disorders,” “psychopathology,” “work-life balance,” “mental health services,” “ADHD,” “autism,”

“intellectual disability,” and “neurodevelopmental disorders.”

We also manually screened reference lists of included studies and performed additional targeted searches using Google Scholar.

Eligibility criteria

- **Inclusion:** Original research and reviews addressing climate change, migration, artificial intelligence, generational change, neurodevelopmental disorders, and mental health in the context of psychiatric training.
- **Exclusion:** Editorials, letters, commentaries without empirical data and studies unrelated to psychiatric education or specialist training.

Screening process

Our search yielded several hundred records across databases. After removing duplicates and applying predefined inclusion criteria, we retained a focused set of articles for in-depth analysis. Given the narrative nature of this review, selection prioritized thematic relevance over exhaustive quantification. The number of articles per subtheme varied according to the scope and relevance of available literature.

Results

Psychiatric training and work-life balance at the intersection of generational and systemic demands

Work-life balance is a crucial topic from a socio-cultural, economic and political perspective and is investigated differently depending on context [71, 102]. Despite the complexity of the topic, research on work-life balance in psychiatric specialty training has so far mainly focused on the phenomenon of burnout, which can be described as a complication resulting from a poor work-life balance [21, 108].

Furthermore current research on work-life balance in the medical field seldomly addresses aspects particularly relevant to younger generations such as sustainability, empowerment, feedback culture and self-actualization even though these topics are of increasing importance in medical training and drive the transformation of professional identities within the healthcare system [70, 75, 89, 98]. The existing structures and traditional concepts lead to growing, seemingly irreconcilable tensions, between older and younger generations of doctors [31, 71].

The younger generations’ desire to balance their professional and personal lives collides with the challenges facing today’s healthcare systems such as increasing economic pressure, chronic staff shortages and rising demand for medical services due to an aging demographic [86]. This discrepancy between individual needs

and systemic demands holds significant potential for conflict. Moreover, junior physicians are confronted with a professional self-image based largely on tradition and often defensively upheld by older colleagues [98]. Apparent differences pertaining to preferences in lifestyle, career expectations and the self-image of doctors further complicate the development of a system of residency training based on intergenerational respect and sensitivities [93]. Fundamental needs in specialist training such as reliable leadership, clear responsibilities, transparency, and emotional safety are often unmet [91]. Consequently, many trainees report feeling overwhelmed, overworked, exhausted and dissatisfied, further exacerbated by impromptu strategies (“proxy methods” and limited autonomy, ultimately contributing to phenomena like the so-called brain drain [71, 86].

Psychiatric training and climate change

At the 2023 DGPPN congress, climate change and the concept of planetary health were chosen as overarching themes, and a comprehensive position paper published afterward called for the integration of climate-related content into psychiatry training curricula [43]. The position paper further emphasized that strengthening empowerment within psychiatric care systems requires adequate climate-related mental health literacy, since clinicians need a solid understanding of how climate change influences mental health in order to exercise autonomy and make informed clinical decisions. However, questions remain about how these desiderata should be implemented and how the acquisition of climate-related competencies should be assessed [50]. To our knowledge, only one study to date has examined how climate-related knowledge could be integrated into psychiatric specialty training [101].

Current scientific studies primarily investigate how climate change both directly and indirectly affects mental health [22] with children, adolescents, elderly people and patients with chronic conditions being especially vulnerable [3, 6, 17, 24, 38, 84]. Furthermore individuals with pre-existing psychiatric disorders appear to be particularly at risk, which is reflected in an increased demand for emergency room consultations during heatwaves [94, 97].

An overall increase in psychiatric emergencies presenting at emergency departments has also been reported [5, 23, 54].

Additional studies found that the prevalence of affective disorders, organic psychiatric disorders, schizophrenia, and neurotic/anxiety disorders increases with high temperatures and during heatwaves [94].

Air pollution has already been linked to depression in many studies, and there is growing evidence of its impact on developmental disorders such as ADHD and autism

[34, 109]. Moreover an increase in suicides and overall mortality has been documented among identified risk groups [44, 110].

Temperature rises and air pollution likely affect mental health through various mechanisms. During prenatal development for example, air pollution can impair placental function, reducing the fetus's supply of oxygen and nutrients. Additionally changes in DNA methylation have been observed [1] and it has also been noted that there seems to be a link between oxidative stress and developmental disorders [59, 80]. Postnatally, exposure to air pollutants is associated with cognitive deficits, increased hyperactivity, and attention problems throughout childhood development [95]. Temperature increases trigger physiological adaptations such as alterations in blood flow or serotonin levels in the brain [15, 39, 61, 77]. Cognitive and behavioral changes such as an increase in feelings of aggression or perceived stress have also been documented [40]. People with psychiatric disorders who take psychotropic medication may be especially vulnerable due to impaired thermoregulation—reduced sweating and thirst sensation [63]. Drought, wildfires, and flooding particularly threaten agricultural regions and contribute to psychological stress. Crop failures and loss of arable land undermine livelihoods and contribute to displacement and migration [49]. On a psychological level, terms like eco-anxiety and eco-distress are increasingly coming to the forefront. These are not yet fully defined pathologies and primarily affect children and adolescents [14, 27]. Climate anxiety is a complex phenomenon and is often based on constructive or practical fear [16]. Although painful and burdensome, it is rational and does not imply mental illness.

In threatening and uncertain situations like the climate crisis, this reaction can be seen as practical fear, as it often motivates people to change their behavior. However, since the climate crisis manifests on many levels and lacks a clear solution, this fear can easily become overwhelming. Concern about climate change is accompanied by young people's perception that their future is under threat and that decisions relevant to climate change are delayed and often not made consistently at the political level. This leads to a sense of betrayal and neglect, which affects intergenerational interactions [45]. Studies often show that younger age groups are more concerned about climate change than older ones [82]. Age-related differences have been found in beliefs about the reality, causes and impacts of climate change, with older individuals more likely to hold climate-skeptical views than younger people [72]. Overall, the results of the most relevant studies suggest that, while there may not be major differences in climate-related beliefs, there is likely a generational divide in risk perception and emotional responses to climate change [87]. Younger generations, compared

to older ones, tend to perceive and process negative emotions such as fear, guilt, and outrage more intensely — a tendency further amplified by predominantly negative media coverage. Older people, by contrast, focus more on positive stimuli and are more effective at filtering out negative ones [20]. Despite climate change being one of the most urgent and serious global issues, people often perceive it as distant and impersonal [99]. Many older adults, for example, believe that climate change does not pose a serious threat within their lifetime [74]. Age seems to play a crucial role in the subjective perception of threats. A longitudinal study conducted in the UK over three years found that emotional responses to climate change showed the greatest generational differences [72]. These examples clearly illustrate that climate change is experienced differently across generations. Climate movements such as *Fridays for Future* and evidence of varying ways different generations process the issue point to an emerging intergenerational conflict.

So far, little is known about the extent to which climate change-related knowledge is integrated into psychiatric specialist training. It also remains unclear whether there are generational differences among physicians in the perception of climate-related phenomena.

One survey has shown that some clinicians express interest in further education and research on the topic [101]. However, other than demands for improving education and research, no concrete steps have been taken so far (DGPPN).

Psychiatric training and artificial intelligence

Artificial intelligence (AI) is increasingly influencing psychiatric research and clinical practice.

AI represents one of the most significant innovations of the twenty-first century and is being progressively integrated into research, clinical care, and medical education across various specialties. This trend is reflected in a notable rise in related scientific publications over the past decade [58, 106].

In the context of intergenerational crises, AI offers promising tools for addressing complex mental health challenges ranging from early detection of psychiatric disorders to personalized treatment approaches [65]. However, these advances also raise important ethical concerns regarding patient privacy, algorithmic bias and the integration of AI into existing psychiatric frameworks [105]. Expectations surrounding AI also vary significantly across generations. Although there are no known studies that specifically examine psychiatrists, it is well documented that attitudes toward AI in medicine differ widely between younger and older physicians [53]. Interestingly, older physicians tend to express more positive attitudes toward the use of AI in everyday clinical practice, despite having a more limited understanding of

the technology [42]. AI has shown substantial potential in improving early detection rates for psychiatric conditions by using machine learning algorithms to identify subtle patterns in patient data. For instance, predictive models analyzing language use are becoming increasingly accurate at identifying individuals at ultra-high risk for psychosis, thereby helping clinicians intervene earlier [10, 29]. Moreover, AI plays a key role in personalizing treatment strategies by processing large datasets to recommend individualized interventions, thereby enhancing treatment efficacy across diverse age groups [35]. AI is also revolutionizing psychiatric research. It enables complex big-data analyses and hypothesis generation that were virtually impossible for most research institutions even 10–15 years ago. Machine learning techniques are being employed to extract diagnostic and predictive patterns from sources such as neuroimaging, sensor data and genomic datasets. Tools like *OpenEvidence*® facilitate literature reviews by providing fast and structured access to scientific publications. Similarly, AI supports academic writing through tools such as *BioRender*® for figure creation and software that assists in drafting manuscripts [58].

To date, no studies have specifically explored generational differences in attitudes or perspectives on artificial intelligence within the psychiatric context. However, various studies have indicated a general openness to integrating AI-related content into medical education [18, 19, 33, 96].

How this integration could be realized within psychiatric training curricula has not yet been clearly defined by professional societies. An initial study on the implementation of AI-assisted teaching methods showed promising results including increased confidence among users in their clinical work [4, 25, 65].

Psychiatric training, transcultural psychiatry and migration

Migration is a significant consequence of climate change, but it is also closely linked to political persecution, armed conflicts, and economic hardship. It is one of the central challenges of the twenty-first century and has a far-reaching impact both directly and indirectly on psychiatric and psychotherapeutic practice [41].

In most Western European countries, one in four people has a migration background, and this number is steadily rising [36].

Culture can be defined as a complex encompassing inherited experiences, ideas and values, which prescribe social order and behavioral norms. It functions as a system of orientation that enables smooth and effective interaction, cooperation, and communication within a group, allowing for the prediction of others' behavior [68]. Culture is shaped by ethnicity, "race," skin color/

appearance, religion, and gender [11]. Both individuals and institutions are imbued with culture.

In psychiatry, there is a distinct culture. In clinical interaction, three cultures meet:

1. **The patient's cultural background** influences how psychiatric disorders—such as depression, anxiety, psychosis, organic conditions, and trauma-related disorders—are presented symptomatically [46, 57]. These culturally shaped presentations affect clinicians' ability to recognize, diagnose, and treat mental disorders. Cultural factors also significantly influence adherence to therapy, illness behavior, coping strategies, treatment response, rehabilitation and recovery. Hofmann and Hinton [48] reviewed culturally bound expressions of anxiety disorders. For example, in Cambodian culture, good health is associated with the free flow of a wind-like substance called *khyal* through straw-like structures in the body. A “blockage” of this wind may have catastrophic results. Cold and numb extremities may signal such blockage, leading to a panic attack and a fear of losing a limb. Conversely, an overly upward flow could stress neck vessels and cause catastrophic consequences. Due to differences in ethnophysiology patients from this culture may therefore not express symptoms typical in the Western world like the feeling akin to having a heart attack, but symptoms related more to the neck area, thus making a proper diagnosis more difficult.
2. **The clinician's own culture** including values, behaviors, life experiences and biases also affects interaction. Examples include evidence-based medicine and patient autonomy. In Western medical systems, physicians emphasize individual autonomy, open communication, and evidence-based diagnostics. Reference systems like ICD-10 or DSM-5 guide the diagnostic process, but they may not adequately account for culturally distinct expressions. Culturally sensitive medicine requires reflection and adaptation of evidence-based diagnostics to the patient's cultural expression.
3. **Cultural differences in general group behavior:** Western cultures tend to be more individualistic, while Eastern and African cultures are more collectivist [66]. Individualistic societies stress personal advancement and self-reliance, whereas collectivist societies prioritize extended-family loyalty, group harmony, and shared identity. Individuals from collectivist backgrounds may resist the Western concept of patient autonomy, preferring inclusion of family in decision-making. Culturally sensitive care means

understanding autonomy differently, in a dignified and context-aware way that respects the individual's social embedding.

In addition to cultural factors, ethno-biological aspects are gaining importance in psychopharmacology [9]. Geographic variations exist in CYP enzyme metabolizer variants, notably CYP2D6, which is especially relevant in psychiatry [67, 69].

Psychosocial and migration-specific factors also significantly influence the prevalence, symptoms and development of mental illness among migrants [12, 51, 56]. Migration is now considered one of the key vulnerability factors for psychosis risk [13, 47]. Post-migration stressors such as loss of cultural identity and acculturative stress are particularly impactful [60, 92]. Additional risk factors include low socioeconomic status, bullying, discrimination, exposure to violence, limited social support, belonging to a sexual minority and migration into areas with low ethnic density [13, 47].

As early as 2015, the European Psychiatric Association's “Guidance on Cultural Competence Training” called for better integration of transcultural psychiatry into specialist training curricula [81], a point explicitly reiterated in the Canadian guidelines of 2021 [55]. Yet, many assistant physicians in psychiatry report feeling poorly prepared by older generations [79, 100].

Transcultural psychiatry has thus far not been standardized. In Austria, for example, the training curriculum for psychiatry and psychotherapy does not even mention it. In Germany and Switzerland, some transcultural elements have been included, but mandatory rotations in transcultural clinics are still absent. In Italy, there are no direct references to transcultural psychiatry in training curricula, though some institutions may offer optional courses [103].

A qualitative study among trainee doctors in India and the U.S. found great interest in migration-related psychiatry training but few had received formal instruction [73, 100]. A clear gap exists between demand, clinical reality and available education for the current generation of psychiatrists in training.

Psychiatric training and neurodevelopmental disorders

Our literature review indicates that ADHD, autism spectrum disorders (ASD), and intellectual disabilities receive limited attention in adult psychiatry training curricula [7, 104]. The literature also highlights that these disorders frequently persist into adulthood—longitudinal studies estimate that approximately 40–55% of individuals with childhood ADHD continue to meet diagnostic criteria in adulthood [32]. Additionally, the interplay between neurodevelopmental disorders and psychosocial vulnerabilities, such as bullying, is well documented. Autistic

individuals, in particular, face high rates of victimization and bullying across the lifespan, which significantly heightens their risk for depression, anxiety, social withdrawal and other psychiatric conditions [2, 30, 83]

In Germany, adult psychiatry training does not include a mandatory rotation in Child & Adolescent Psychiatry (KJP); neurodevelopmental disorders remain primarily within the separate KJP specialty—potentially reinforcing diagnostic and therapeutic gaps in adult mental health practice [104]

Internationally, some training bodies—such as the UK's Royal College of Psychiatrists—have begun integrating neurodevelopmental content (ADHD, ASD, intellectual disabilities) into revised curricula and ongoing CPD requirements. However, survey data show that psychiatrists consistently report lower confidence in diagnosing and managing ASD than ADHD, reflecting notable disparities in practical preparedness despite formal training exposure [26, 28].

Discussion

Psychiatry is being reshaped by intersecting global drivers—climate change, artificial intelligence (AI), and migration—alongside generational shifts among trainees. While empathy and evidence-based experiential knowledge remain the foundation of practice, this foundation must now evolve to support competencies and learning cultures fit for a rapidly changing world [8].

Despite clear awareness of relevance and urgency, these themes have not been systematically or structurally integrated into postgraduate curricula or continuing professional development. Early implementations show enthusiasm but uneven confidence and patchy coverage—particularly for sustainability, AI literacy, transcultural competence and adult neurodevelopmental care [43, 65, 81].

A concrete and comprehensive picture thus emerges: policy signals exist, but translation into routine training remains incomplete. In the climate domain, professional pledges and briefings have defined the emergency and its mental-health implications, yet trainees frequently report limited mastery without targeted teaching, supervised practice and assessment [43, 65, 76].

In the digital domain, pilot curricula in digital psychiatry can raise confidence in app appraisal and digital therapeutics, while scoping reviews in medical education reveal fragmented frameworks and a shortage of trained faculty—underscoring the need for specialty-specific competency maps and faculty development [58, 65].

In the transcultural domain, European guidance consistently urges cultural and structural competence to reduce disparities for migrants and minorities, yet implementation remains uneven, pointing to the necessity of practical tools (e.g., skilled interpreter use, cultural

formulation, navigation of entitlements) embedded in routine assessments [9, 81].

Finally, life-course neurodevelopmental conditions remain under-addressed in adult training, despite longitudinal syntheses demonstrating ~40–55% persistence of childhood ADHD into adulthood and repeated calls to strengthen adult ASD services and transition pathways between CAMHS and AMHS [32, 85].

In line with the idea that “progress needs setbacks,” a deliberate pause can enable perspective-taking, error recognition, and customization of innovations—an insight that resonates with psychodynamic views of crisis as a catalyst for developmental shifts and inner growth. Educational reforms often falter not because the content is wrong but because ingrained experiential patterns—hierarchical orientation, efficiency biases, or the hidden curriculum—silently shape what is learnable in practice [78].

If these patterns remain unexamined, reflection risks becoming rhetoric without structural impact. Confrontation, peer dialogue, and co-production with service-user educators can help surface and rebalance power, moving training from declarative exposure to operational competence [88, 90].

New demands—work-life balance, practice-oriented teaching formats and the attrition or retirement of experienced colleagues—require continuous curriculum adaptation. Generational studies and trainee surveys repeatedly show that long working hours, limited supervision, and low job control are associated with burnout, while supportive supervision and autonomy buffer risk and enhance resilience [52, 64].

Large national data [62] highlight overload and staffing deficits across hospitals—nearly half of employed physicians feel frequently overloaded and 28% consider leaving—conditions that can blunt educational uptake unless addressed at organizational and policy levels [62]. Feelings of being overwhelmed predict emotional exhaustion and downstream anxiety/depressive symptoms, which undermine learning and patient care; structured guidance and reliable supervision become crucial anchors that restore orientation and a sense of security [37, 52, 107].

It remains crucial that the implementation process depends not only on theoretical understanding but focuses on re-designing learning environments. To make sustainability, AI, transcultural competence and life-course neurodevelopmental care truly learnable, programs should embed supervised, case-based practice and assess competence in authentic clinical encounters.

This entails: integrating eco-distress formulations and sustainable prescribing into routine supervision and workplace-based assessments [43, 76], defining AI literacy specific to psychiatry (safety, bias, governance, appraisal of algorithms and apps) and examining it via

OSCEs with AI-mediated scenarios [65], operationalizing transcultural competence through required simulations and rotations (interpreter use, cultural formulation, entitlement navigation), paired with service-level quality-improvement on access [9], and setting minimum supervised experiences in adult ADHD/ASD assessment, medication management, and transitions, aligned with persistence data that justify adult capacity [32, 85].

Real change arises from confrontation and exchange—the everyday conversations among trainees, supervisors, service-user educators and partners in primary care, social services, and public health. Structural and personal attitudes shape training reality and, by extension, how diagnosis and therapy evolve in a pluralistic society. Co-produced teaching not only distributes expertise more evenly but also improves learner well-being and engagement when power asymmetries are named and ethically managed [88, 90]. In practice, the older generation's role is less about supplying answers and more about creating spaces for experience—psychologically safe settings where guided participation allows younger colleagues to test, fail, reflect and consolidate. The younger generation's role is to step into these spaces with curiosity and accountability.

Limitations

This review adopts a narrative integrative approach, combining peer-reviewed literature with insights from expert symposia and congress discussions. While this enriches conceptual depth, it also introduces subjectivity, as interpretations partly reflect consensus-building in these forums. Despite prioritizing high-quality sources (including meta-analyses and guidance from leading journals), thematic selection rather than exhaustive systematic screening may entail selection bias. Finally, the evidence base for some innovations (e.g., AI literacy curricula, sustainability training) remains emergent and largely qualitative, limiting generalizability and requiring ongoing updates as new empirical data and policy frameworks evolve.

Generalizability

Although much of the cited literature derives from UK/European and North American contexts (EPA Guidance, Academic Psychiatry/JMIR/BJPsych/European Psychiatry, DGPPN Guidance), the drivers (climate risk, AI proliferation, migration flows) and training challenges (cultural competence, adult NDD care, hidden curriculum, supervision) recur globally. The proposed transformations—competency-based content, supervised practice, psychological safety, and co-production—are adaptable to local epidemiology, legal frameworks, and service configurations while maintaining core principles of sustainability, equity, and life-course care [43, 81].

Key takeaways

Updating psychiatry for the next decade is as much a relational project as a content project: beyond adding climate, AI, transcultural, and life-course neurodevelopmental topics, training must transform *how* we teach—toward supervised, co-produced, psychologically safe, and competency-assessed learning that reliably translates into confident, equitable everyday practice.

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Author contributions

I specified this section within our cover letter.

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Data availability

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Declarations

Ethics approval and consent to participate

This work does not involve any issues regarding ethical standards.

Competing interests

The authors declare no competing interests

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